

Dear Professor Marc & Selection Team,

I am writing to apply for the EngD on developing an **interactive reverse logistics map** for circular construction. The topic caught my attention for a simple reason: if people can see materials, logistics and choices in one place, reuse becomes much more likely.

In my master's thesis, **Agile Project Management in the Construction Industry**, I studied how Agile can help construction teams work better together, manage risk, make decisions and use resources more carefully. When I finished, I felt something was still missing. Better delivery and less waste on site only cover part of sustainability. I started asking myself what happens to materials when a building reaches the end of its life.

That path took me into circular construction and reverse logistics. I also read earlier work in the field, including Professor Voordijk's research on recovering building elements for reuse. One point stayed with me: reuse needs coordination from the donor building to the target building, not only new technology. In practice, information about availability, quality, ownership, location, transport and reuse options is often missing or spread across many people. When that happens, new materials feel safer. I have also looked at Circular Matching Platforms, which try to connect material supply from one project with demand in another. An interactive reverse logistics map can go a step further: show what is available, help match donor supply with target demand even when timing, quality and cost are uncertain, and let users compare reuse on cost, CO₂ and circularity. I do not want to build another tool that sits unused. I want to help build something people will actually use.

My recent jobs have prepared me for this kind of work. At Aug.e I worked on digital twins for smart buildings. We used a digital model to understand and improve how a real building performs. At Hive CPQ I managed large product datasets from manufacturers and turned complex information into simple tools that non-technical users could work with. At iostiring I worked with a construction partner on a circular construction hypothesis: how Agile project management can bring circular practices into real projects. We did not reach a full pilot. Working on a (de)construction pilot with partners like Dura Vermeer and TNO would let me take that step, using my civil and environmental engineering background and GIS skills. I have been on a career break to care for my newborns, and I kept learning about this field during that time.

I would like to do this EngD at the University of Twente. I like that the programme mixes advanced study with a real design project in industry. For me, that means learning reverse-logistics modelling in more depth, getting better at GIS for matching materials, and working closely with partners: start from real use cases, test the map in a pilot, and improve it with the people who will use it. I can relocate to the Netherlands.

Thank you for reading my application. I would be glad to talk further.

Yours sincerely,

Pooja Awasare